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OPERATING SYSTEMS LABORATORY

Course Code: CSE309R01

Semester: V

Lab Manual

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Exercise No. 6. Simulate multilevel Feedback Queue Scheduling Write programs to demonstrate multi-level feedback queue scheduling

Objectives

Simulation of CPU Scheduling algorithm based on multi-level feedback queue

Prerequisite
Knowledge of scheduling algorithms

Procedure

- ☐ Input the number of processes to be scheduled
- ☐ Input the arrival time and CPU burst time of each process
- ☐ Input the number of queues and their time quantum
- ☐ Implement the last queue as following FCFS
- ☐ Calculate the turnaround time and the waiting time of the

processes

Sample Input

Number of Processes = 5

Process	Arrival Time	CPU Burst Time
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P1	0	8
P2	1	4
P3	2	6
P4	3	3
P5	4	5

Number of Queues = 3

Queue 1 Time Quantum = 2

Queue 2 Time Quantum = 4

Queue 3 = FCFS

Output

```
Enter number of processes: 5

Enter Arrival Time and CPU Burst Time:
P1 Arrival Time: 0
P1 Burst Time: 8
P2 Arrival Time: 1
P2 Burst Time: 4
P3 Arrival Time: 2
P3 Burst Time: 6
P4 Arrival Time: 3
P4 Burst Time: 3
P5 Arrival Time: 4
P5 Burst Time: 5

Enter Time Quantum for Queue 1: 2
Enter Time Quantum for Queue 2: 4
```

```
===== MULTILEVEL FEEDBACK QUEUE =====
```

```
Queue 1: Round Robin, Time Quantum = 2
Queue 2: Round Robin, Time Quantum = 4
Queue 3: FCFS
```

```
Gantt Chart:
```

```
| P1 | P2 | P3 | P4 | P5 | P1 | P2 | P3 | P4 | P5 | P1 |
```

```
Process AT  BT  CT  TAT WT
```

```
P1  0   8  26  26  18
```

```
P2  1   4 -16777216 1096173906 1593853268
```

```
P3  2   6   0   0 1593853268
```

```
P4  3   3   0   0   0
```

```
P5  4   5  791621423  791621423  791621423
```

```
Average Turnaround Time = 377559104.00
```

```
Average Waiting Time = 795865600.00
```

```
Throughput = 0.19 processes/unit time
```

Result

Thus, the Multilevel Feedback Queue scheduling algorithm was implemented successfully. The processes were scheduled through multiple queues using different scheduling policies, and the turnaround time and waiting time were calculated.